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Lateral Open Boundary Conditions:

The Key to Reliable Ocean Forecasts

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Lateral Open Boundary Conditions:

The Key to Reliable Ocean Forecasts

- Introduction
- Practical consideration
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- Good Practices
- Conclusions

Introduction

The primary goal of using a Limited Area Model is to accurately simulate ocean dynamics within a specific region.

The accuracy and reliability of a LAM depend heavily on the proper functioning of the lateral open boundaries, which serve as gateways for the exchange of information between the model and the surrounding ocean.

The issue you must face arise from:

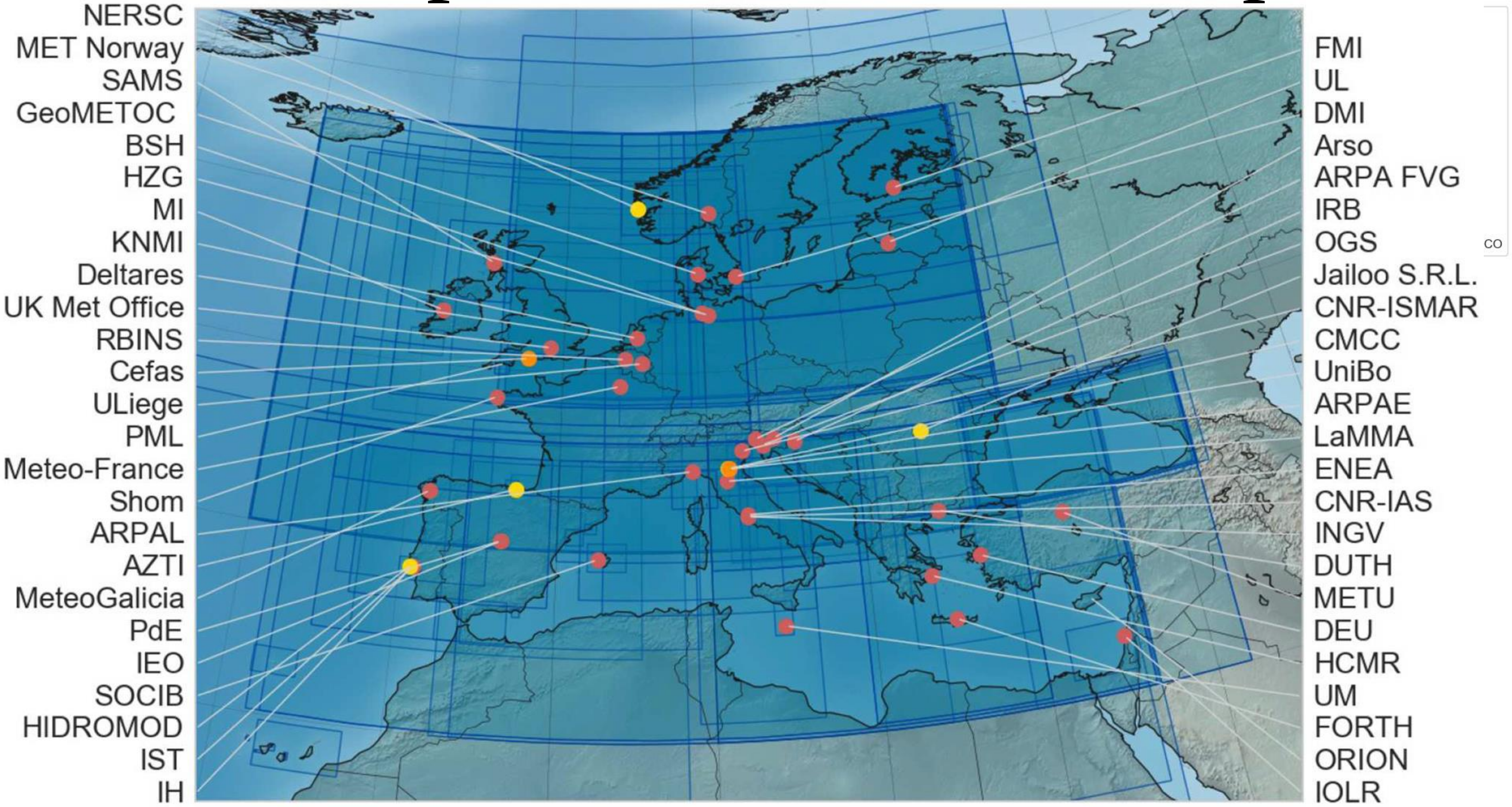
- Inconsistency between the “external” and “internal” solutions.
- Numerical stability.

Introduction

Is it relevant?

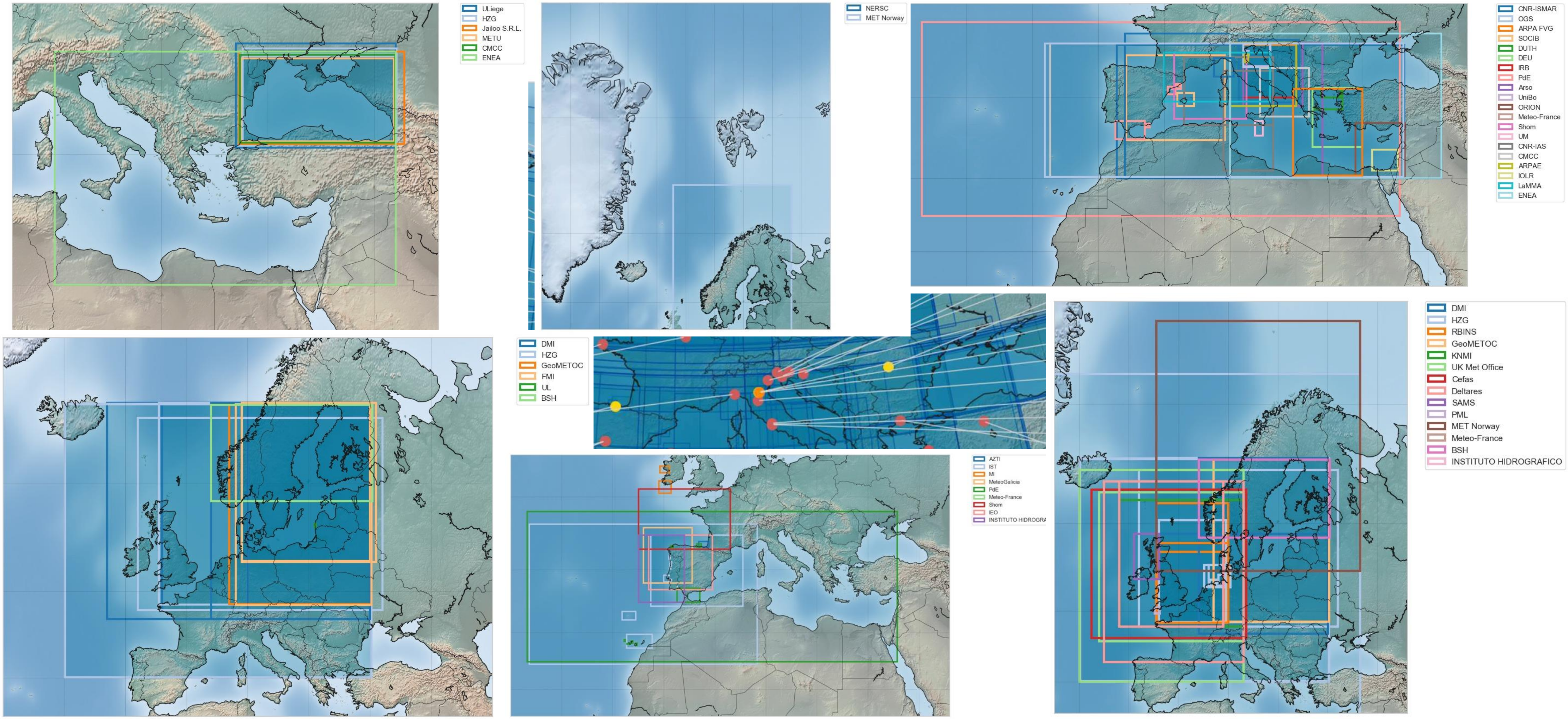
Introduction

Example of LAMs in Europe



Capet A, Fernández V, She J, Dabrowski T, Umgiesser G, Staneva J, Mészáros L, Campuzano F, Ursella L, Nolan G and El Serafy G (2020) Operational Modeling Capacity in European Seas—An EuroGOOS Perspective and Recommendations for Improvement. *Front. Mar. Sci.* 7:129. doi: 10.3389/fmars.2020.00129

Introduction



Practical Consideration

As the exact solution is lacking, practical implementations of open boundary conditions have been sought.

The effectiveness of the specification adopted must be evaluated with respect to the characteristics of the problem at hand and with regards to the spatial and temporal scales of interest.

Practical Consideration

Even if the specification of prognostic variables on the open boundary is ill-posed in the pure mathematical sense, errors due to this ill-posedness may be insignificant for the flow space and time scales of interest.

In this case, for any practical purpose, the boundary conditions specified may be considered viable.

If, on the other hand, the boundary conditions generate or reflect waves that propagate inside the domain, or if they do not effectively transmit information into the interior, or allow information to exit, then they are not considered satisfactory.

Available Options

Many open boundary conditions have been proposed in the literature.

Some of them are simple schemes that nudge the prognostic variables to a reference state within a specified region in the proximity of the open boundary with an a priori relaxation time.

Others are based on approximations of the primitive equations, thereby providing a local solution based on “reduced physics”.

Available Options

The relaxation/nudging scheme may be considered the simplest solution to the LOBC problem.

It consists of the addition of a Newtonian relaxation term to the model governing equations. The most drastic way to do this is to impose:

$$\theta = \theta^{\text{ext}}$$

Available Options

Nudging Layer: a zone in proximity of the lateral open boundary conditions where the solution of the child model is relaxed toward the solution of the parent model to increase coherency/consistency between the two “estimates”

$$\lambda(\theta - \theta_{\mathbf{R}})$$

Sponge Layer: a zone in proximity of the lateral open boundary conditions where viscosity/diffusivity is artificially increased in order to suppress small scale features resolved by the child model and not present in the parent model

Available Options

The advective conditions are such that the normal velocity at the boundary from the nested (regional) model is used to advect out of the domain the field:

$$\frac{\partial \theta}{\partial t} + V_{\mathrm{n}} \frac{\partial \theta}{\partial n} = 0$$

Thus, external data are advected inward at inflow boundary points and the interior solution is advected outward by the regional model velocities.

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Available Options

Popular open boundary conditions are derived from the Sommerfield radiation equation (1949). If the interior solutions approaching the open boundary propagate through it in a wave-like form:

$$\frac{\partial \theta}{\partial t} + C\frac{\partial \theta}{\partial n} = 0 \qquad \frac{\partial \theta}{\partial t} + C_x\frac{\partial \theta}{\partial x} + C_y\frac{\partial \theta}{\partial y} = 0$$

An approach to estimate *C* was proposed by Orlanski (1976)/Raymond and Kuo (1984):

$$C = -\frac{\partial \theta}{\partial t}\left(\frac{\partial \theta}{\partial n}\right)^{-1}$$

$$C_x = -\frac{\partial \theta}{\partial t}\frac{\partial \theta/\partial x}{(\partial \theta/\partial x)^2 + (\partial \theta/\partial y)^2}$$

$$C_y = -\frac{\partial \theta}{\partial t}\frac{\partial \theta/\partial y}{(\partial \theta/\partial x)^2 + (\partial \theta/\partial y)^2}$$

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Available Options

Miyakoda and Rosati (1977) suggested prescribing the external information at inflow points and using the “Orlanski Radiation” at outflow. The “Modified Orlanski Radiation”. To avoid inconsistency at the boundary points where the flow field switches between the two regimes, Marchesiello *et al*. (2001) proposed to add a nudging term:

$$\frac{\partial \theta}{\partial t} + C_x \frac{\partial \theta}{\partial x} + C_y \frac{\partial \theta}{\partial y} = -\frac{1}{\tau}(\theta - \theta^{\text{ext}})$$

The authors suggest and test the idea of using different time scales for the nudging term depending on the inflow–outflow regime.

Available Options

An interesting and useful solution for the open boundary conditions was proposed by Flather (1976), prescribing a relationship between elevation and currents at the open boundary arising from a combination of the continuity equation with a radiation condition. This condition can be classified as a radiation condition.

$$\nabla \cdot [(H_c + \eta_c)\vec{V}_{\mathrm{BT}c}] + \frac{\partial \eta_c}{\partial t} = \nabla \cdot [(H_f + \eta_f)\vec{V}_{\mathrm{BT}f}] + \frac{\partial \eta_f}{\partial t}$$

$$\frac{\partial \eta}{\partial t} = -\nabla \cdot (\vec{C}\eta)$$

$$V^n_{\mathrm{BT}f} = \frac{H_c + \eta_c}{H_f + \eta_f} V^n_{\mathrm{BT}c} - \frac{C^n}{H_f + \eta_f}(\eta_c - \eta_f) \qquad V^n_{\mathrm{BT}f} = V^n_{\mathrm{BT}c} - \frac{\sqrt{gH}}{H}(\eta_c - \eta_f)$$

Good practices

A widely accepted principle among researchers dealing with Lateral Open Boundary Conditions was encapsulated by the phrases:

“The further away, the better”

“Distance is key”

“More distance is better”

Good practices

Still, we need to ensure the robustness of our LAM

- Matching the bathymetries in proximity of the lateral open boundaries, reduces discontinuities in the discretized space.
- If possible, we can build the LAM grid considering the parent model grid
- Combining a nudging and a sponge layer will increase numerical stability and reduce the differences between the two models

Further we want to preserve "relevant" quantities when interpolating from the parent to the child grid:

- The concept of integral constraint was introduced by Zavatarelli and Pinardi (2003)

$$\int_{x_2}^{x_1} \int_{-H_{\text{nested}}}^{\eta_{\text{nested}}} V_{\text{nested}} dz\, dx = \int_{x_2}^{x_1} \int_{-H_{\text{nesting}}}^{\eta_{\text{nesting}}} V_{\text{orig}} dz\, dx \;,$$

Good practices

Always remember the practical considerations.

If errors due to ill-posedness are insignificant for the flow space and time scales of interest, the boundary conditions specified can be considered viable.

If the boundary conditions generate or reflect waves that propagate inside the domain, or if they do not effectively transmit information into the interior, or allow information to exit, they cannot considered satisfactory.

Conclusions

- An exact analytical solution does not exist.
- Due to differences in scale and complexity, there will always be an inconsistency between parent and child model solutions.
- Remember rule N.1:“The further away, the better”.
- A large variety of options are available in literature. Select the method that yields the most accurate results.
- Ensure geometrical consistency, preserve the relevant information, damp small-scale features near the boundary, and maintain continuous communication between parent and child models to minimize inconsistencies, are a good starting point.

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